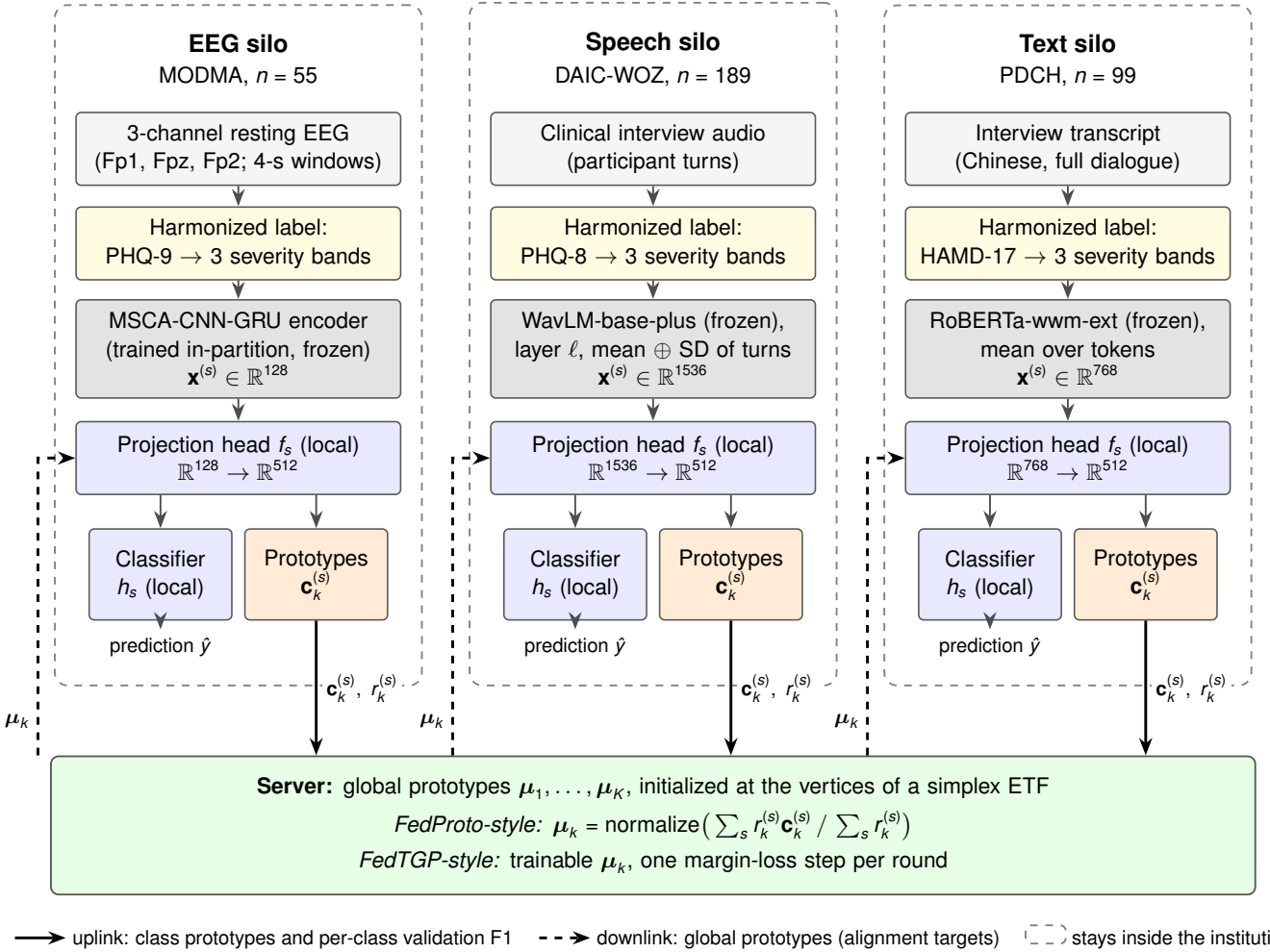

EEG silo
MODMA, n = 55
3-channel resting EEG (Fp1, Fpz, Fp2; 4-s windows)
Harmonized label: PHQ-9 → 3 severity bands
MSCA-CNN-GRU encoder (trained in-partition, frozen)
x(s) ∈ ℝ128
Projection head fs (local)
ℝ128 → ℝ512
Classifier hs (local)
Prototypes ck(s)
prediction ŷ
ck(s), rk(s)
μk
Speech silo
DAIC-WOZ, n = 189
Clinical interview audio (participant turns)
Harmonized label: PHQ-8 → 3 severity bands
WavLM-base-plus (frozen), layer ℓ, mean ⊕ SD of turns
x(s) ∈ ℝ1536
Projection head fs (local)
ℝ1536 → ℝ512
Classifier hs (local)
Prototypes ck(s)
prediction ŷ
ck(s), rk(s)
μk
Text silo
PDCH, n = 99
Interview transcript (Chinese, full dialogue)
Harmonized label: HAMD-17 → 3 severity bands
RoBERTa-wwm-ext (frozen), mean over tokens
x(s) ∈ ℝ768
Projection head fs (local)
ℝ768 → ℝ512
Classifier hs (local)
Prototypes ck(s)
prediction ŷ
ck(s), rk(s)
μk
Server: global prototypes μ1, . . . , μK, initialized at the vertices of a simplex ETF
FedProto-style: μk = normalize(∑s rk(s)ck(s) / ∑s rk(s))
FedTGP-style: trainable μk, one margin-loss step per round
uplink: class prototypes and per-class validation F1
downlink: global prototypes (alignment targets)
stays inside the institution